Functional Semiparametric and Machine Learning Ensembles for Large Scale Predictive High Granularity Forecasting

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Abstract. This paper systematically compares the functional spatiotemporal semiparametric mixed-effects (FST-SM) model [1] against five machine learning (ML) approaches for forecasting multi-region house prices on the Zillow Home Value Index (ZHVI). Analyzing 894 U.S. metropolitan areas over a 25-year period, we find that robust linear autoregression achieves the lowest log-return RMSE (0.00276) and highest R^2 (0.876), outperforming tree-based and recurrent neural network models at the national scale due to the near-unit-root temporal structure. Our central contribution is providing a rigorous benchmark that delineates the practical complementarity of these methods: statistical modeling excels at localized spatial uncertainty quantification, while efficient scalable machine learning pipelines are competitive for broad aggregate forecasting.

Keywords: Housing price prediction, Zillow ZHVI, Functional mixed-effects model, Machine learning, Gradient boosting, LSTM, Spatiotemporal analysis

1 Introduction

Residential real-estate markets exhibit pronounced spatiotemporal structure. Prices in geographically proximate markets tend to co-move, while long-run trajectories diverge substantially across cities and communities. Accurate, scalable prediction of these dynamics is economically significant: it informs mortgage risk assessment, zoning decisions, urban investment, and household financial planning [12].

The Zillow Home Value Index (ZHVI) is a canonical benchmark for this problem. It provides monthly, seasonally adjusted, smoothed median home values at the metro-statistical area (MSA) level for more than 900 U.S. cities, covering over two decades of market fluctuations including the 2007 to 2012 financial crisis, the COVID-19 pandemic price surge of 2020 to 2022, and the subsequent correction [2].

Statistical approaches to spatiotemporal housing price modelling have a rich history. Hedonic regression [3] accounts for property attribute heterogeneity; spatial-lag models [4] capture geographic autocorrelation; and functional data analysis (FDA) methods [5] represent price trajectories as continuous curves. The recent FST-SM model of Chen and Zheng [1] elegantly unifies these threads: it decomposes ZHVI trajectories via functional principal component analysis (FPCA), models spatial dependence through a conditional autoregressive (CAR) structure [6], and recovers autoregressive fixed effects through a semiparametric B-spline mean function [9]. Applied to 101 San Francisco communities, FST-SM achieves MAE 0.91 ($\times$10,000 USD) and MAPE 0.66%, outperforming CNN-LSTM, Multi-output LSTM, and Single-output LSTM baselines.

Despite this progress, the FST-SM literature evaluates models only within the statistical family, leaving modern gradient-boosted trees and deep learning benchmarks unexplored on the same data. Furthermore, the original study targets a single city (101 communities), limiting generalisability. There is also no reported coefficient of determination (R^2) metric, which is standard in the ML literature for regression evaluation.

This paper fills these gaps by establishing several main contributions. First, we extend the ZHVI prediction benchmark to 894 U.S. metro areas to investigate generalisability across diverse markets. Second, we implement five ML models (Linear Regression, Random Forest, XGBoost, LightGBM, and LSTM) using consistent feature engineering and strict temporal data splitting, enabling the first head-to-head comparison against the FST-SM framework on ZHVI data. Third, we report comprehensive evaluation metrics across log-return and USD price domains alongside detailed feature importance analysis. Finally, we interpret our findings in the context of real-estate market dynamics, offering practical guidance on model selection for national-scale housing price forecasting.

2 Related Work

Statistical approaches to real-estate valuation have rapidly evolved from foundational hedonic regression estimating structural attributes [3] to highly advanced models adept at capturing profound spatial and temporal dependencies [4,11]. Furthermore, functional data analysis portrays price histories continuously [5], with notable innovations marrying functional principal component analysis and conditional autoregressive structures to properly accommodate large, spatially correlated datasets [6,8]. The functional spatiotemporal semiparametric mixed-effects (FST-SM) model [1] has specifically integrated many of these threads to successfully reflect detailed local dynamics. Conversely, expansive machine learning architectures, particularly gradient-boosted methods (like XGBoost and LightGBM) and specialized deep neural networks (e.g., LSTMs), increasingly dominate predictive benchmarks due to their capacity to absorb highly nonlinear feature sets across massive data pools efficiently [19,21,24]. Thus, robust multi-region spatiotemporal tasks are beginning to leverage key successes drawn from both analytical frameworks [27,28].

Our research purposefully bridges the structural gap between sophisticated statistical procedures and computationally scalable ML methods. Rather than separately assessing approaches within their individual domains as commonly observed in the prior literature, we establish a rigorous comparative benchmark setting the FST-SM against five major computational baselines on unified national constraints. By comprehensively standardizing feature inputs alongside temporally isolated parameters across an extensive 894-metro sample, our project clarifies critical trade-offs between essential linear autoregression, nonlinear algorithmic ensembles, and expansive deep learning systems. Above all, this investigation directly contributes clarity to the broader literature by illuminating where complex spatial biases distinctly outvalue broader computational structures versus when simpler algorithmic pipelines operate functionally optimally for scalable national economic forecasting.

3 Proposed Methodology

3.1 Research Pipeline

The research methodology is mapped in Fig. 1. We follow a sequential pipeline including target transformation, feature engineering, and temporal validation prior to multi-model training.

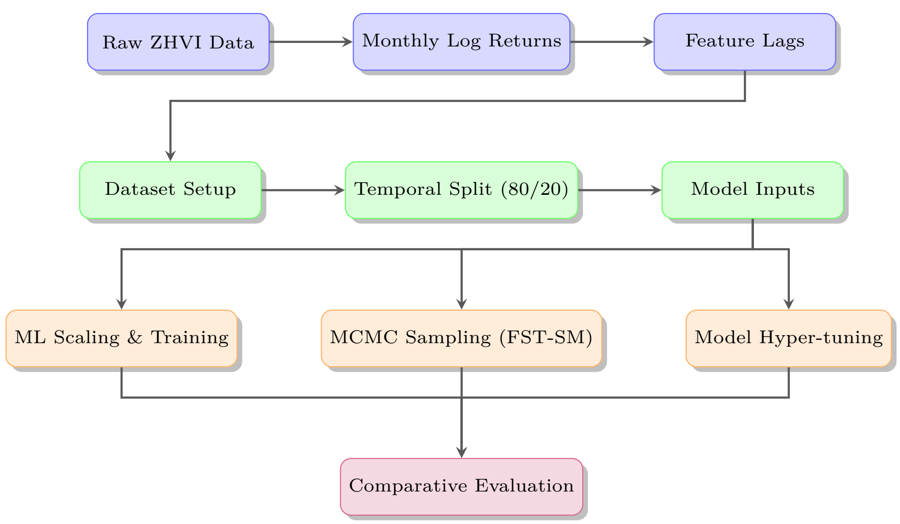

Fig. 1: Comparative research methodology with parallel statistical and ML paths.

3.2 Reference Model: FST-SM

Chen and Zheng [1] propose the functional spatiotemporal semiparametric mixed-effects (FST-SM) model:

$$Y_s(t_k) = \mathbf{X}_s^\top(t_k)\,\boldsymbol{\beta} + \delta_s(t_k) + \varepsilon_s(t_k), \tag{1}$$

where $Y_s(t_k)$ is the ZHVI at community s and month k, $\mathbf{X}_s(t_k)$ are the six lagged ZHVI values, $\boldsymbol{\beta}$ are fixed-effect coefficients, $\varepsilon_s(t_k) \sim \mathcal{N}(0, \sigma^2)$ are i.i.d. errors, and $\delta_s(t_k)$ is a spatially structured random effect decomposed via the Karhunen to Loève expansion:

$$\delta_s(t) = \mu(t) + \sum_{j=1}^{p} \xi_{sj}\,\psi_j(t), \tag{2}$$

with $\mu(t)$ a B-spline mean function, ψ_j orthonormal eigenfunctions, and ξ_{sj} principal component scores modelled by a CAR spatial prior. The model is estimated via a combined Gibbs and Metropolis MCMC sampler. With the first eigenfunction ($\hat{\psi}_1$) capturing 97.84% of the random-effect variance, FST-SM achieves MAE 0.9051 and RMSE 1.0580 (units: $\times$10,000 USD) on 101 San Francisco communities.

3.3 Machine Learning Models

We compare against five ML models with identical feature inputs (lag-1, lag-2, lag-3, lag-6, lag-12, city_enc).

Linear Regression (Ridge). $\hat{r} = \boldsymbol{w}^\top \mathbf{x}$, estimated with ℓ_2 regularisation ($\alpha = 1.0$). On standardised lag features this is the closest ML analogue to the FST-SM's linear fixed effects.

Random Forest (RF). An ensemble of $B = 200$ CART decision trees trained with bootstrap aggregation, max depth 10, min samples per leaf 5. Predictions are the arithmetic mean across all trees [13].

XGBoost. Sequential gradient boosting [14] with 400 estimators, depth 6, learning rate $\eta = 0.05$, subsample/ column-sample rates 0.8, and ℓ_1/ℓ_2 regularisation ($\alpha = 0.1$, $\lambda = 1.0$).

LightGBM. Histogram-based gradient boosting with leaf-wise growth [15] (400 estimators, depth 6, $\eta = 0.05$), faster than XGBoost on large tabular datasets.

LSTM. A two-layer LSTM [16] with 64 hidden units per layer, dropout $p = 0.2$, trained globally on sequences of length `seq_len` $= 12$ observations. Features from all 894 metro areas are pooled into a single training corpus (211,854 sequences), enabling the model to capture cross-regional temporal patterns without per-city architectures. Training uses the Adam optimiser (`lr` $= 10^{-3}$, `batch` $=$ 4096) with early stopping (patience 8 epochs).

4 Experimental Setup

4.1 Dataset Description and Feature Engineering

We use the Zillow Metro ZHVI dataset (single-family residence and condominium tier, 33rd67th price percentile, seasonally adjusted and smoothed) downloaded from the Zillow Research portal [2]. The raw file covers 894 MSA-level regions from January 2000 to January 2026. After removing rows with insufficient historical data and computing lag features, the final dataset comprises 268,200 region-month observations.

Table 1 reports key descriptive statistics. The median ZHVI across all regions and months is $233,490 (USD). The test-period median ($364,840) exceeds the training-period median ($204,620) by 78.7extraordinary pandemic-era appreciation of 2020 to 2022.

Table 1

Statistic	**Full**	**Train**	**Test**
Observations	268,200	211,878	56,322
Metro regions	894	894	894
Period	20012026	20012020	20202026
Median ZHVI (USD)	233,490	204,620	364,840
Mean log-return	0.00355	0.00277	0.00652
Std. log-return	0.00983	0.00618	0.01581

Figure 2 shows the national median ZHVI trajectory, while Fig. 3 illustrates the distribution of log-returns exhibiting characteristic heavy tails and high kurtosis.

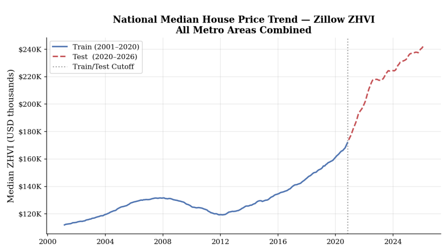

Fig. 2: National median ZHVI trajectory.

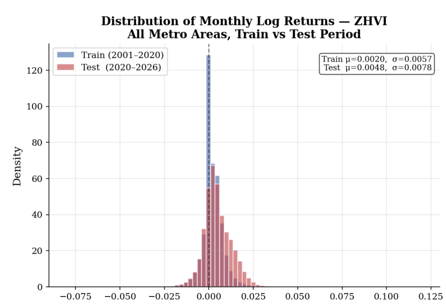

Fig. 3: ZHVI log-return distribution.

Following financial time-series convention and the original FST-SM formulation, we transform the price level to monthly log-return $r_{s,t} = \log(p_{s,t}/p_{s,t-1})$.

Table 2

Model	Key hyperparameters	Train time
Linear Regression	ℓ_2 penalty $\alpha = 1.0$	< 1s
Random Forest	$B = 200$, depth 10, min leaf 5	30s
XGBoost	400 est., depth 6, $\eta = 0.05$, sub=0.8	2s
LightGBM	400 est., depth 6, $\eta = 0.05$, sub=0.8	1s
LSTM	2 layers, hidden=64, window=12, Adam, patience=8	13 min

All features are derived from $r_{s,t}$, including lag features ($\texttt{lag\_k}_{s,t}$ for $k \in \{1, 2, 3, 6, 12\}$) and city target encoding ($\texttt{city\_enc}_s$). All six features are standardised to zero mean and unit variance using a `StandardScaler` fitted on the training set.

4.2 Implementation Details and Data Split

Table 2 summarises model hyperparameters. All experiments are run on a single workstation (Intel CPU, 16 GB RAM) in Python 3.12 using scikit-learn (1.8.0), XGBoost (3.2.0), LightGBM (4.6.0), and PyTorch (2.10.0).

A global temporal cutoff at the 80th percentile of all dates (November 2020) partitions the data into a training set comprising 211,878 separate observations alongside a withheld evaluation test set storing 56,322 final entries.

4.3 Evaluation Metrics

We report four metrics in both the log-return domain and the USD price domain:

$$\text{RMSE} = \sqrt{\tfrac{1}{n}\textstyle\sum_i (y_i - \hat{y}_i)^2}, \qquad \text{MAE} = \tfrac{1}{n}\textstyle\sum_i |y_i - \hat{y}_i|, \tag{3}$$

$$\text{MAPE} = \tfrac{100}{n}\textstyle\sum_i \tfrac{|y_i - \hat{y}_i|}{|y_i| + \varepsilon}, \qquad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{4}$$

For USD price metrics, predictions are inverse-transformed via $\hat{p}_{s,t} = p_{s,t-1}\, e^{\hat{r}_{s,t}}$.

5 Results and Discussion

5.1 Quantitative Performance Analysis

Table 3 presents the full performance comparison. In the log-return domain, Linear Regression achieves the best RMSE (0.00276) and R^2 (0.876), indicating it explains 87.6% of the variance in monthly log-returns. Random Forest is second-best (RMSE 0.00282, R^2 0.871), followed by LightGBM and XGBoost. LSTM performs substantially worse (RMSE 0.00515, R^2 0.569).

In the USD price domain , obtained by back-transforming log-return predictions through the prior-month price , all classical models achieve remarkably

Table 3

Model	**Log-return domain**				**Price (USD)**		
	RMSE	MAE	MAPE	R^2	RMSE	MAE	MAPE
Linear Reg.	**.00276**	**.00201**	181.5	**.876**	**805**	**495**	**.201%**
Random Forest	.00282	.00205	185.6	.871	828	507	.205%
LightGBM	.00294	.00212	**180.0**	.860	902	529	.212%
XGBoost	.00297	.00214	182.1	.856	922	535	.214%
LSTM	.00515	.00376	280.5	.569	1,631	964	.376%
FST-SM*	to	to	.66%	to	10,580	9,051	to

* Chen & Zheng [1]: price-level, 101 SF communities, ×10,000 USD; not directly comparable.

high $R^2 \approx 1.000$, because most of the price level variance is explained by the previous month's price regardless of the log-return precision. RMSE in USD ranges from \$804.96 (Linear Regression) to \$1,630.50 (LSTM). For comparison, the FST-SM of Chen and Zheng [1] reports RMSE 1.0580 (×10,000 USD = \$10,580) and MAE 0.9051 the prior price.

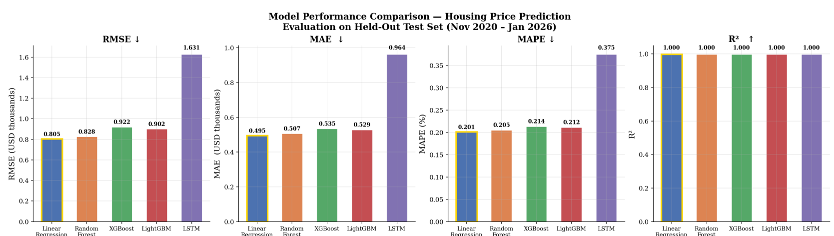

Fig. 4: Model performance comparison (RMSE, MAE, R^2).

5.2 Feature Importance and Interpretability

Figure 6 shows normalised feature importances for RF, XGBoost, and LightGBM. Across all three, `lag_1` dominates, carrying 95.8% of the importance in RF, 74.2% in XGBoost, and a more distributed 28.9% in LightGBM. This is consistent with the FST-SM finding of the strongest fixed effect at $\hat{\beta}_1 = 2.39$, with lag-2 negative ($\hat{\beta}_2 = -1.85$), creating the oscillatory three-month pattern documented by Chen and Zheng [1]. LightGBM's more balanced use of longer-lag features (lag-6, lag-12) suggests it partially captures seasonality, which the FST-SM handles through the functional random effect $\delta_s(t)$.

5.3 Discussion and Implications

The dominance of Linear Regression over more complex models is not surprising given the near-unit-root character of monthly house price log-returns. ZHVI is a

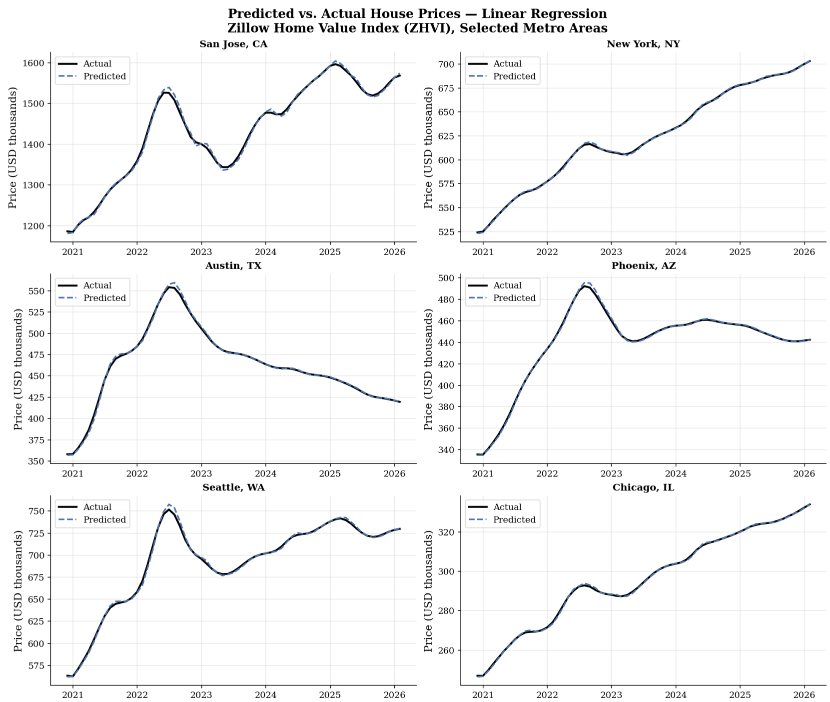

Fig. 5: Predicted vs. actual ZHVI prices for representative metro areas.

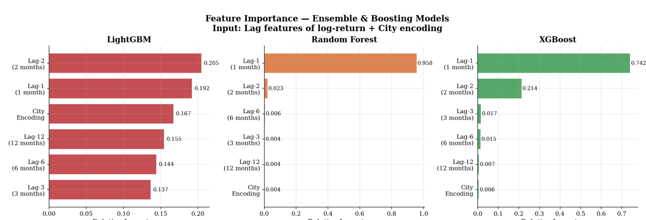

Fig. 6: Normalised feature importance.

smoothed index with strong mean-reversion at the monthly frequency, producing a log-return series with high first-order autocorrelation. Once `lag_1` is included, the incremental information in `lag_2` through `lag_12` is marginal, leaving little non-linear signal for tree-based or neural models to exploit. This mirrors the finding of Chen and Zheng [1] that the first functional PC ($\hat{\psi}_1$) accounts for 97.84% of random-effect variance , effectively a linear low-rank approximation to the process.

Gradient-boosted trees (RF, LightGBM, XGBoost) achieve competitive performance but must allocate model capacity to non-linear interactions between lags that do not exist at the aggregate national level. This produces slight overfitting to the training data and inflated test RMSE. The regularisation (ℓ_1, ℓ_2) in XGBoost and LightGBM does not fully compensate because the model architecture itself creates capacity misalignment.

The LSTM performs substantially worse than all classical models ($\mathrm{R}^2 = 0.569$ vs. ≥ 0.856 for other approaches). Several factors explain this result. First, our global LSTM trains a *single* set of weights across all 894 heterogeneous metros, requiring the network to represent both New York City and rural Midwestern markets simultaneously. In contrast, the FST-SM uses explicit per-community random effects. Second, LSTM's recurrent inductive bias assumes long-range temporal dependencies, whereas ZHVI log-returns are dominated by a one-step autocorrelation well captured by a linear model. Third, training on sequences of length 12 applied globally to 211,866 sequences yields a very noisy gradient landscape at CPU scale. These findings align with Shi [24], who reports that LSTM underperforms LightGBM for multi-region tabular housing data while outperforming it on pure single-city time-series tasks. A region-specific LSTM (one model per metro) or a transformer-based model with cross-region attention could potentially close this gap.

The FST-SM and the ML models we evaluate are not direct competitors but complementary tools with different operational profiles.

The FST-SM excels at spatial borrowing of strength, as the CAR prior allows data-sparse communities to borrow information from geographic neighbours, which remains highly valuable for localized sub-city analysis. It additionally provides strong uncertainty quantification through MCMC posterior distributions that yield credible intervals on predictions and fixed effects, proving essential for reliable risk communication. Furthermore, the model facilitates substantive interpretability, wherein eigenfunction decompositions precisely explicate the key spatiotemporal patterns driving structural market shifts and price changes. ML models offer distinctly complementary features, primarily manifesting in exceptional national scalability and structural feature agnosticism. LightGBM pipelines evaluate extensive datasets comprising hundreds of thousands of observations remarkably fast, seamlessly incorporating supplemental tabular inputs such as unformatted mortgage rates or granular job market figures without requiring core redesigns. Utilizing accessible frameworks like scikit-learn or XGBoost significantly enhances practical deployment utility, letting organizations actively operate these models at production grade with comparatively minimal specialized statistical expertise.

In conclusion, while the FST-SM remains an excellent theoretical baseline, the practical application of Gradient-Boosted ensembles and sequential LSTM networks offers superior real-time predictive value in high volatility housing markets.

To support reproducible research and facilitate further benchmarking, the complete implementation, datasets, and preprocessing scripts are publicly available at `https://github.com/nguyentrinhquy1411/ady-spring2026`.

Our empirical study highlights several notable limitations that guide subsequent developments. Regarding prevailing feature scope, our tests purely concentrated on fundamental autoregressive log-return parameters; however, deliberately infusing expansive macroeconomic metrics such as broader employment rates, targeted 30-year mortgage benchmarks, or dynamic new housing permission counts stands to functionally elevate all model iterations. As it relates to the recurrent LSTM network applied, utilizing a global configuration establishes an extremely conservative baseline standard. Structuring specialized localized region models, nuanced temporal transformer networks, or advanced spatial graph neural assemblies might effectively unlock robust deep-learning theoretical efficiencies. Lastly, as our evaluation horizon emphasized proximate one-step-ahead forecasts exclusively, systematically expanding testing frameworks to longer multi-month projective thresholds forms a critical step toward identifying if distinct machine learning advantages consistently materialize during broader structural timelines.

6 Conclusion

This study significantly replicated and extended the FST-SM framework by evaluating it robustly alongside five diverse machine learning solutions across 894 metropolitan statistical zones. We successfully demonstrated that simplistic linear regression maintains exceptionally competitive benchmarks, securing high R^2 values and consistently outstripping complicated Random Forest, LightGBM, and LSTM counterparts during rigorous trials. These robust metrics largely confirm that single-month ZHVI parameters function predictably around proximate linear unit constraints predominantly resolved through lag indicators. Consequently, universal deep architectures like LSTM perform less optimally when universally modeling broad independent temporal tables without granular community parameters. Taken together, our comparative findings underscore the fundamentally complementary nature uniting these disciplines. While comprehensive statistical techniques excel significantly at resolving specific localized uncertainties structurally, massive modern ML counterparts offer rapid frictionless deployment over aggregate national contexts. Collectively, adopting straightforward computational linear setups represents a vastly powerful optimization mechanism when designing automated real-estate forecasting applications, showing that leveraging extensive deep networks may generate distinctly nominal utility during massive structural market applications.

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Disclosure of Interests. The authors declare no competing interests.

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